7. Short notes ANY FOUR

(i) Recovery , Recrystallization and grain growth

(ii) Hot and cold deformation

(iii) Dendritic segregation

(iv) Precipitation hardening

(v) T-T-T diagrams 5x4

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. ..120MEC 1116

B.E. II (M)

2 **Mat. Tech.**

SECOND B.E.(MECHANICAL ENGINEERING)

III SEMESTER

EXAMINATION - 2019

ME-202 A: MATERIALS TECHNOLOGY (M)

Time - Three Hours

Maximum Marks - 100

Note:- (1) Answer any **FIVE** questions.

(2) All questions carry equal marks.

(3) All the parts of a question should be attempted in continuation.

(4) Give neat sketches wherever necessary.

1. Describe with examples

(i) Unit cell

(ii) Atomic Packing Factor

(iii) Co-ordination number

(iv) Crystal Symmetry

(v) Atomic radius

Derive A.P.F., co-ordination number and atomic radius for all structures (10)

(b) Explain the system of Miller indices for planes and directions of crystal lattice. (10)

2. (a) What are the defects and imperfections in crystal? Describe them with neat sketches. (10)

(b) Differentiate between edge dislocation and screw dislocation. Con (10)

3. (a) Explain the difference between slipping and twinning. How does twinning occur in metals? Name and explain two types of twins. 10

(b) What is critical resolved shear stress? On what factors does it depend? 5

(c) What is effect of cold working on the mechanical properties of metals? 5

4. (a) What do you understand by a solid solution? Explain with neat sketches the subsitutional solid solution and interstitial solid solution. Give two example of each in alloy system. 10

(b) Draw the phase diagram for a binary system showing complete solubility in liquid and solid state. 10

5. (a) Explain briefly the following heat treatment operations.

(i) Annealing

(ii) Normalising

(iii) Tempering

(iv) Hardening

(v) Nitriding 10

(b) Describe martensite transformation in steel. Name the quenching media that are used for heat treatment. Discuss their merits and demerits. 10

6. (a) Discuss Modified Iron Carbon Diagram. Explain their importance in practical applications. 10

(b) Describe the process of decomposition of Austenite to the following

(i) Pearlite

(ii) Ferrite

(iii) Cementite

(iv) Bainite

(v) Martensite 10

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. ...16UMEC1118.

B.E. II (M)
2

Mat. Tech.

Second B.E. Mechanical Engineering
III SEMESTER
EXAMINATION-2018
ME-202 A: MATERIALS TECHNOLOGY
(M)

Time - Three Hours
Maximum Marks - 100

Note :- (1) Answer any FIVE questions.
(2) All Questions Carry Equal marks.
(3) All the parts of a question should be attempted in Continuation.
(4) Give neat sketches wherever necessary.

1. (a) Draw a cooling curve for a binary alloy, and hence. explain the mechanism of Crystallization ie solidification showing micro-structures at various stages. 10

(b) Explain the following:

(i) Spacelattice and its charateristics.

(ii) Atomic Packing factor, derive APF for FCC. crystal structure 5+5

2. (a) With neat sketches explain the following imperfections in Crystalline solids

(i) Interstitial defect

(ii) Edge dislocation 5+5

Write the Causes of their Occurance

(b) Define the term miller indices for plane. Given are the miller indices for the planes (112), (110) & (212) draw the planes in a simple cubic unit cell. 4+6

3. (a) Draw a Stress Strain Curve for a ductile material and explain various nomenclatures of the curve.

(b) Define the term fracture. Sketch and explain various stages of a ductile fracture. 4+6

4. (a) Define the term toughness of a material sketch and explain the V-notch method (Charpy method) for determining the impact strength of a metal. 4+6

(b) Define the term hardness of a metal. Sketch and explain the Rockwell method for determining hardness of a metal. 4+6

5. (a) Define the terms Phase and Phase diagram. 5

(b) Two metals A and B having their melting temperatures as 1450°c and 1100°C respectively, and showing mutual solubility in liquid and Solid State formining homogeneous mixture.

(i) Draw the phase diagram of the alloy assuming suitable scale for percentage composition by weight and temperature in degree centigrade.

(ii) Determine the chemical composition of alloy at 1300 C.

(iii) Relative amount of liquid and solid phase present in two phase region at 1300°C and for composition of 60 Percent A. 5+5+5

6. (a) For what pupose hardening of steel is done. Write the process steps for hardening of steel. Show the micro structure changes occuring during process. 10

(b) Define the term annealing. Explain the full annealing process for a 0.2% Carbon Steel sketch microstructure Changes Occuring during Process. 10

7. (a) Differentiate between the following

(i) Medium and high Carbon Steels.

(ii) Ferritic and Austenitic Stainless Steels. 5+5

(b) Write notes on following

(i) Normalising of steel and microstructure Changes for hypo eutectoid steels.

(ii) Nitriding method of case harding of steels. 5+5

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 16 MEC 3 8 5 15

B.E.II (M)

2

Mat.Tech.

SECOND B.E. MECHANICAL ENGINEERING
III SEMESTER EXAMINATION - 2017
ME-202 A : MATERIALS TECHNOLOGY (M)

Time Allowed- Three Hours

Maximum Marks : 100

Note: (1) Attempt any **five** questions.

(2) All questions carry equal marks.

(3) All the parts of a question should be attempted together.

(4) Give neat sketches wherever necessary.

(5) All parts of a question should be answered in succession.

1. (a) Describe the crystal structures of metallic elements. What characteristics of the atomic structure of metals accounts for their relative high thermal and electric conductivity. (10)

(b) Describe with example:-

(i) Unit Cell

(ii) Atomic Packing Factor

(iii) Miller Indices

(iv) Co-ordination number (10)

2. (a) Sketch following planes and directions

(i) (1,1,2)

(ii) (1,0,1)

(iii) (2,2,2)

(iv) (4,0,0)

(v) (0,0,1) (10)

(b) How may atoms per square millimeter are there on the (1,0,0) plane of lead. Assume interatomic distance to be 3.499 A°. (10)

3. (a) What are the point, line and surface imperfections found in solid materials. Illustrate these imperfections with suitable sketches. (10)

(b) Differentiate between edge dislocation and screw dislocation with help of suitable sketches. (10)

4. (a) Explain the term slip and twin. How does slip occur? Explain slip directions and slip planes with diagrams. (10)

(b) What is the effect of cold working on the mechanical properties of metals and alloys?

Discuss the changes in properties when a severely cold worked metal is annealed at successively higher temperature. (10)

5. (a) What do you understand by a solid solution? Explain with neat sketches the subsitutional solid solution and interstitial solid solution. Give two examples of each in the alloy system. (10)

(b) What is the object of heat treatment? List the various heat treatment process. Describe process annealing and full annealing. (10)

6. (a) What is meant by thermoplastics and thermosetting plastic? Distinguish between these clearly. State their properties and industrial applications. (10)

(b) Draw iron carbon equillibirium diagram and show their salient features. Indicate the significance of this diagram for heat treatment of steel. (10)

7. Write short notes **ANY four** -

(a) Nano and smart materials

(b) Bearing materials

(c) Mechanical and electrical properties of materials

(d) Bouschinger effect

(e) Gibb's Phase rule (5x4)

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. ..15.ME.. ....

B.E. II (M)

2 **Mat. Tech.**

Second B.E. Examination 2016

(Mechanical Engineering)

(III-Semester Examination Scheme)

ME-202 A : MATERIALS TECHNOLOGY (M)

Time - Three Hours

Maximum Marks - 60

Note : (1) Attempt any **Five** Questions.

(2) All Questions carry equal marks.

(3) All the parts of a question should be attempted together.

(4) Give neat sketches wherever necessary.

(5) All parts of a quetion should be answered in succession.

1. (a) Explain the composition of solid illustrating cystals, unit cell, atoms and electrons in it.

(b) Define primitive unit cell. Molecular cystal and coordination number. 6

2. (a) Define atomic packing factor. Obtain its expression for SC. FCC and BCC 6

(b) Derive the expression which relates interplanner spacing. miller indices and dimension of the (a) cubic unit call (b) tetragonal unit cell 6

3. (a) Sketch the following planes and directions 6

(i) (221).

(ii) (101)

(iii) (1.0.0)

(iv) (2.3.4)

(v) (400)

(b) Define the following imperfection in crystal.

(i) Schottky (ii) Frankel (iii) Compositional

(iv) Screw dislocation (v) Edge dislocation 6

4. (a) Plastic deformation is mainly caused by slip. Why so justify? 6

(b) Describe different slip system in various materials. Explain schmidt law of CRSS. 6

5. (a) Explain following mechanical properties.

(i) Elasticity

(ii) Plasticity

(iii) Ductility

(iv) Malleability

(v) Hardress

Explain the mechanism of solidifications. 3+3

(b) Derive Gibb's phase rule. Explain solidification of alloys and dendrite growth. 6

6. (a) Draw iron carbon equillibirium diagram and show their salient features. Indicate the significance of this diagram for heat treatment of steel. 6

(b) Explain the principle of heat treatment. Explain following cyniding. nitriding flame hardening, Induction hardening. 6

7. (a) Write short notes (Any Four) 3×4=12

(i) Compaction and Sintering

(ii) Bearing Materials

(iii) Nano and smart materials

(iv) Precipitation Hardening

(v) Hot and Cold Working

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 13MEC48076

B.E. II (M)

2 **Mat. Tech.**

Second B.E. Examination, 2015

(Mechanical Engineering)

(III-Semester Examination Scheme)

ME–202 A : MATERIALS TECHNOLOGY (M)

Time—Three Hours

Maximum Marks 60

Note :— (1) Attempt any **FIVE** questions.

(2) All questions carry equal marks.

(3) All the parts of a question should be attempted together.

(4) Give neat sketches wherever necessary.

(5) All parts of a question should be answered in succession.

1. (a) Define atomic packing factor. Calculate its value for simple cubic, body centered cube and face centered cube structure. 6

(b) What is meant by powder metallurgy ? State the advantage and limitation of powder metallurgy. 6

2. (a) What do you mean by "Miller indices" ? State the important features of Miller indices. 6

(b) Explain the different types of molecular structure of polymers. 6

3. (a) What are the typical alloys of copper used in engineering ? Discuss briefly their composition and uses. 6

(b) Differentiate between the following :

(i) Slip and twinning

(ii) Edge dislocation and screw dislocation. 6

4. (a) State the effects of adding alloying elements to steel :

Nickel, Chromium, Tungsten, Silicon. 6

(b) What is an equilibrium diagram ? Give the classification of equilibrium diagram. Explain any one case with neat diagram. 6

5. (a) What is TTT diagram ? How the TTT diagram is prepared ? Explain. 6

(b) Explain the theory of tempering. What are the effects of tempering on the mechanical properties of steel ? 6

6. (a) Define the term heat treatment and state its objectives.. Enumerate various heat treatment processes with the help of temperature – % carbon diagram. 6

(b) Explain the recovery, recrystallisation and grain growth when a cold worked metal heated subsequently. 6

7. Write short notes on any **FOUR** :

(a) Burgers vector

(b) Solid solution

(c) Carburising

(d) Important Aluminium alloys and their uses

(e) Strain hardening

(f) Classification, advantages and application of ceramics.

$3 \times 4 = 12$

7. Write short notes on **ANY FOUR:**

(a) Recovery, Recrystallization and grain growth.

(b) Phase diagram-components and phases.

(c) Austempering and Martempering

(d) Applications of Nano materials, smart materials and super conductors.

(e) Eutectic, Eutectoid and Peritectic reactions. (3x4=12)

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. . .3200056. .

B.E. II (M)

2

Mat. Tech.

Second B.E. Examination, 2014
(Mechanical Engineering)
(III - Semester Examination Scheme)

ME - 203 A : MATERIALS TECHNOLOGY (M)

Time : Three Hours
Maximum Marks : 60

Note : (1) Attempt any **FIVE** questions
(2) All questions carry equal marks.
(3) All the parts of a question should be attempted together.
(4) Give neat sketches wherever necessary.
(5) All parts of a question should be answered in succession.

1.(a) Distinguish between amorphous and crystalline substances. In what patterns do formation of crystals take place? Explain with neat sketches. (6)

(b) Explain following in various crystalline structures

(a) Atomic radius

(b) Atomic packing factor

(c) Coordination number (6)

2.(a) Explain Miller indices for denoting crystal planes and directions. (6)

(b) Draw (110) and (1,1,1) planes and [1,1,0] and [1,1,1) directions in a simple cubic crystal. What do you infer from these diagrams? (6)

3.(a) Explain the various mechanism for dislocation in plastic deformation with neat sketches. (6)

(b) Distinguish between hot working and cold working. What is the effect of cold working on the mechanical properties of metals and alloys? Explain briefly the term 'Strain hardening.' (6)

4.(a) In carbon-Iron equilibrium diagram explain following :

(a) allotropic transformation

(b) Critical points and critical range

(c) decomposition of Austenite

(b) What do you mean by isothermal transformation? What useful information can be obtained from these curves. Construct TTT diagram to show different phases. (6)

5.(a) Enumerate the objectives of heat treatment of Steel. What are the various processes to obtain desired objective? (6)

(b) Why hardening is essential in few applications? Explain various hardening processes. (6)

6.(a) What is the powder technology technique. Describe briefly the methods of powder forming suitable for this technique? (6)

(b) What do you mean by bearing materials? Briefly discuss the effect of alloying elements in steel. Discuss the applications of polymers and composites. (6)